

SOCIOL 723: Social Statistics II
Week 9, Regression Discontinuity Designs

Wenhao Jiang

Department of Sociology, Fall 2026



Today

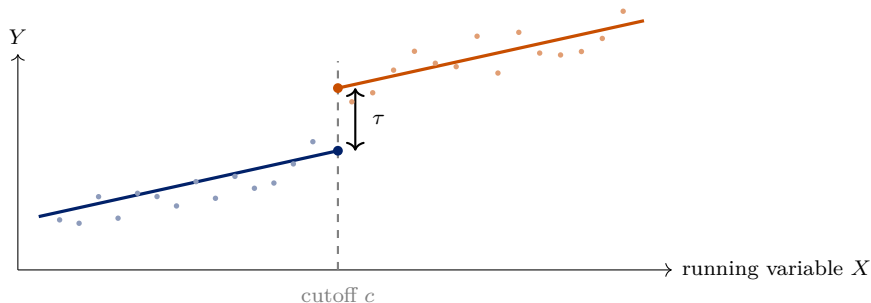
Sharp Regression Discontinuity

Fuzzy RD

Validity

The Idea in One Picture

Many treatments are assigned by a rule: a test score threshold, an income cut-off, a vote share above 50%, an age of eligibility.



Just below and just above the cutoff, units are essentially identical, except that one group got the treatment. The jump in the outcome at c is the effect.

Sharp RD

Setup and Identification (cont.)

A **running variable** (or forcing variable) X_i determines treatment deterministically:

$$D_i = \mathbf{1}\{X_i \geq c\}.$$

Continuity assumption (Hahn, Todd, and van der Klaauw 2001)

$E[Y_i(1) \mid X_i = x]$ and $E[Y_i(0) \mid X_i = x]$ are continuous in x at $x = c$.

Then

$$\tau_{RD} = \lim_{x \downarrow c} E[Y_i \mid X_i = x] - \lim_{x \uparrow c} E[Y_i \mid X_i = x] = E[Y_i(1) - Y_i(0) \mid X_i = c].$$

- Note what is *not* assumed: nothing about the shape of $E[Y \mid X]$ away from the cutoff, and no ignorability anywhere else.

Setup and Identification (cont.)

- **Positivity fails at every x , the cutoff included**: assignment is deterministic, so $\Pr(D = 1 \mid X = x)$ is 0 or 1 and never strictly between. Identification therefore does *not* come from overlap. It comes from **continuity** of $x \mapsto E[Y_i(d) \mid X_i = x]$, which lets the two one-sided limits stand in for the missing counterfactual at c . That is also why the estimand is local to c .

The Estimand Is Local, and That Matters

τ_{RD} is the average effect **for units at the cutoff**. Nothing more.

- A scholarship RD tells you about students at the eligibility margin, not about the strongest or weakest applicants.
- Extrapolating away from c requires a functional-form assumption of exactly the kind RD was designed to avoid.
- Compare with the LATE from Week 7: both are estimands you did not choose, determined by the design rather than by the question. State them plainly.
- In exchange, RD is often the most credible observational design there is. The identifying variation is visible in a scatterplot.

Lee (2008)'s reformulation is helpful: if agents cannot perfectly control X_i , then treatment near the cutoff is *as good as randomly assigned*, and the design is a local experiment.

Estimation: Local Linear Regression

Fit separate linear regressions on each side of the cutoff, using only observations within a bandwidth h , weighted by a kernel:

$$\hat{\tau} = \hat{\mu}_+ - \hat{\mu}_-, \quad (\hat{\mu}_+, \hat{\beta}_+) = \arg \min \sum_{i: c \leq X_i < c+h} K\left(\frac{X_i - c}{h}\right) (Y_i - \mu - \beta(X_i - c))^2,$$

and symmetrically below.

- **Local linear, not global polynomial.** High-order global polynomials give bizarre weights to distant observations and are now strongly discouraged (Gelman and Imbens 2019). Use degree 1 or 2.
- **Triangular kernel** (weights falling linearly to zero at h) is the standard choice and is boundary-optimal.
- Always let the slope differ on the two sides; forcing a common slope imposes a restriction you have no reason to believe.

The Estimator, Written Out

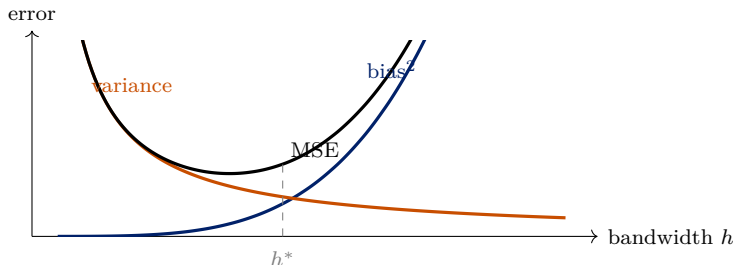
The RD estimate is the treatment coefficient in a **kernel-weighted, fully interacted local linear regression** on the observations within the bandwidth:

$$\min_{\mu, \tau, \beta_-, \beta_+} \sum_{i: |X_i - c| \leq h} K\left(\frac{X_i - c}{h}\right) \left(Y_i - \mu - \tau D_i - \beta_-(X_i - c)(1 - D_i) - \beta_+(X_i - c)D_i\right)^2,$$

with $D_i = \mathbf{1}\{X_i \geq c\}$ and $\hat{\tau}$ the estimate.

- **Centre the running variable at c** , so that the intercept shift τ is evaluated at the cutoff rather than at $X = 0$.
- **Interact the slope with D_i** . Forcing a common slope imposes a restriction you have no reason to believe, and it lets data on one side contaminate the estimate on the other.
- **Triangular kernel** $K(u) = (1 - |u|)_+$ weights observations linearly down to zero at the boundary and is minimax optimal at a boundary point. A uniform kernel is simpler and usually similar.

The Bandwidth Is the Whole Game



- **Wide h :** more data, lower variance, but the linear approximation is worse far from c , bias.
- **Narrow h :** less bias, but few observations, variance.
- Choose h by minimizing estimated MSE (Imbens–Kalyanaraman; the Calonico–Cattaneo–Titiunik version in `rdrobust`).
- **The MSE-optimal bandwidth leaves a bias that invalidates the conventional interval.** Use the CCT *robust bias-corrected* interval, which `rdrobust` reports by default.

Why the Optimal Bandwidth Breaks the Confidence Interval (cont.)

The local linear estimator has an asymptotic expansion

$$\hat{\tau}_h - \tau \approx \underbrace{B h^2}_{\text{bias}} + \underbrace{O_p((nh)^{-1/2})}_{\text{noise}},$$

so squared bias grows like h^4 while variance falls like $1/(nh)$.

- Minimizing MSE gives $h^* \propto n^{-1/5}$, at which the bias and the standard error are **of the same order**.
- Consequently the conventional interval $\hat{\tau}_{h^*} \pm 1.96 \hat{se}$ is *not* centred on τ : it undercovers, sometimes badly.

Why the Optimal Bandwidth Breaks the Confidence Interval (cont.)

- Calonico, Cattaneo, and Titiunik (2014): estimate the bias with a higher-order local polynomial, subtract it, and **widen the interval to account for having estimated the bias**. That is the “robust” row in `rdrobust` output.
- **Report the robust interval**. Reporting the conventional one at an MSE-optimal bandwidth is internally inconsistent.

Covariates, Power, and What to Report (cont.)

- **Covariates are for precision only.** They must be *pre-determined*, they should enter additively (not interacted with D), and the estimate should barely move when you add them. If it moves a lot, the design is in trouble, not improved (Calonico et al. 2019).
- **Effective sample size:** report the number of observations *inside the bandwidth* on each side. An RD with $n = 100,000$ often has 400 observations doing the work, and the standard errors should be read accordingly.
- **Power:** RD is a local method and is inherently data-hungry. The required sample size is a substantial multiple of an experiment's for the same power, often quoted as around $3\times$, though the exact factor depends on the design, because we are extrapolating to a boundary from one side at a time.

Covariates, Power, and What to Report (cont.)

- **Discrete running variables and heaping:** rounded ages, coarse test-score grids, or income bunched at round numbers do not invalidate the design, but they do invalidate the continuous- X asymptotics behind bandwidth selection and bias correction. Use a discrete-score method and say so.

Fuzzy RD

When the Rule Is Not Followed

Often crossing the cutoff changes the *probability* of treatment rather than determining it: eligibility is not take-up.

$$\lim_{x \downarrow c} \Pr(D_i = 1 \mid X_i = x) - \lim_{x \uparrow c} \Pr(D_i = 1 \mid X_i = x) \in (0, 1).$$

The fuzzy RD estimator is a ratio of two sharp RD estimates:

$$\tau_{FRD} = \frac{\text{jump in } Y \text{ at } c}{\text{jump in } D \text{ at } c} = \frac{\lim_{x \downarrow c} E[Y \mid x] - \lim_{x \uparrow c} E[Y \mid x]}{\lim_{x \downarrow c} E[D \mid x] - \lim_{x \uparrow c} E[D \mid x]}.$$

- This is exactly the Wald estimator of Week 7, with $\mathbf{1}\{X \geq c\}$ as the instrument, estimated locally.
- The numerator alone is the **intention-to-treat** effect of crossing the cutoff, and it is worth reporting on its own, it needs no first-stage assumption.

Fuzzy RD Inherits Everything From IV (cont.)

- **Exclusion:** crossing the cutoff must affect Y only through D . If the threshold also triggers something else, a different school, a different caseworker, a label, exclusion fails.
- **Monotonicity:** no one is treated *because* they fell below the cutoff.
- **The estimand is doubly local:** the effect for compliers at the cutoff. Not the ATE, not the ATT, and not the effect for everyone near the cutoff.
- **Weak first stage** is a real risk: if take-up jumps by only a few percentage points, you are dividing by a small noisy number and all the Week 7 pathologies return. Report the size of the jump in D .

Fuzzy RD Inherits Everything From IV (cont.)

Related designs

Regression kink: identification from a change in *slope* rather than level.

Geographic RD: the running variable is distance to a boundary, and the “cutoff” is two-dimensional. **Multi-cutoff RD:** different units face different thresholds, which can be normalized and pooled.

Validity

The Threat: Manipulation

Continuity fails if units can *precisely* control which side of the cutoff they land on, retaking a test until you pass, reporting income just under a threshold, a firm holding hiring just below a regulatory headcount.

- **Imprecise control is fine.** Lee (2008)'s insight is that if agents cannot control X to the last decimal, assignment near the cutoff is as good as random even though agents influence X .
- **Density test** (McCrary 2008; Cattaneo, Jansson, and Ma 2020): is the density of X continuous at c ? A spike on the advantaged side is direct evidence of sorting.
- **Covariate smoothness:** run the RD with *pre-determined covariates* as outcomes. They should show no jump. This is the single most persuasive check, and it is the RD analogue of a balance table.
- **Donut hole:** drop observations immediately around c , where manipulation would concentrate, and see whether the estimate survives.

Other Checks and Common Errors

- **Placebo cutoffs:** estimate at fake thresholds away from c , using only data on one side so the real discontinuity is not picked up. Significant “effects” there mean your procedure finds jumps everywhere.
- **Bandwidth sensitivity:** plot $\hat{\tau}$ against h over a wide range with robust intervals. A result that exists only at one bandwidth is not a result.
- **Do not use a high-order global polynomial** (Gelman and Imbens 2019). The implied weights on distant observations are large, oscillating, and indefensible; estimates wander with the degree.
- **Do not control for post-treatment covariates**, and do not interact covariates with D unless you mean to estimate heterogeneity.
- **Do not report only the conventional interval** at an MSE-optimal bandwidth.

Reading and Reporting an RD

1. **The picture first.** Binned means of Y against X , with the fitted local polynomials and the cutoff. If the jump is not visible, be sceptical of the number.
2. The bandwidth, how it was chosen, and the effective n on each side.
3. Robust bias-corrected confidence intervals.
4. The density test and the covariate-smoothness checks.
5. A bandwidth-sensitivity plot and placebo cutoffs.
6. For fuzzy designs, the size of the jump in treatment take-up, and the intention-to-treat effect alongside.
7. An explicit statement that the estimand is local to the cutoff.

Looking Ahead

- RD buys extraordinary credibility at the cost of a very local estimand. The identifying variation is visible in a scatterplot, and the main threats (sorting, bandwidth dependence) are checkable.
- Note the family resemblance: sharp RD is a local experiment; fuzzy RD is local IV. The tools recombine rather than multiply.
- **Next week:** designs that use variation over *time* rather than across a threshold.
- Note again how the estimand was chosen by the design rather than by us. Saying so plainly is part of reporting an RD honestly.

Problem Set 4

Assigned today, due Monday November 2. Covers Weeks 7 and 9.

References (cont.)

- Calonico, Sebastian, Matias D. Cattaneo, and Rocío Titiunik. 2014. “Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs.” *Econometrica* 82(6): 2295–2326.
- Calonico, Sebastian, Matias D. Cattaneo, Max H. Farrell, and Rocío Titiunik. 2019. “Regression Discontinuity Designs Using Covariates.” *Review of Economics and Statistics* 101(3): 442–451.
- Cattaneo, Matias D., Michael Jansson, and Xinwei Ma. 2020. “Simple Local Polynomial Density Estimators.” *Journal of the American Statistical Association* 115(531): 1449–1455.
- Gelman, Andrew and Guido Imbens. 2019. “Why High-Order Polynomials Should Not Be Used in Regression Discontinuity Designs.” *Journal of Business & Economic Statistics* 37(3): 447–456.
- Hahn, Jinyong, Petra Todd, and Wilbert van der Klaauw. 2001. “Identification and Estimation of Treatment Effects with a Regression-Discontinuity Design.” *Econometrica* 69(1): 201–209.
- Imbens, Guido W. and Thomas Lemieux. 2008. “Regression Discontinuity Designs: A Guide to Practice.” *Journal of Econometrics* 142(2): 615–635.
- Imbens, Guido W. and Karthik Kalyanaraman. 2012. “Optimal Bandwidth Choice for the Regression Discontinuity Estimator.” *Review of Economic Studies* 79(3): 933–959.

References (cont.)

- Lee, David S. 2008. “Randomized Experiments from Non-Random Selection in U.S. House Elections.” *Journal of Econometrics* 142(2): 675–697.
- Lee, David S. and Thomas Lemieux. 2010. “Regression Discontinuity Designs in Economics.” *Journal of Economic Literature* 48(2): 281–355.
- McCrary, Justin. 2008. “Manipulation of the Running Variable in the Regression Discontinuity Design: A Density Test.” *Journal of Econometrics* 142(2): 698–714.